Relatorio de Segmentacao de Poros

EDS / CBS / QEMSCAN - Analise Morfometrica PARTISAN

Imagem analisada : S1.png (Grão 3)

Data de geracao : 30/03/2026 18:58

Porosidade total : 29.155%

Poros analisados : 70

Gerado por: pore_segmentation (Ilastik 1.4 + PARTISAN v2.0)

Parametros de Configuracao

```
ilastik.ilastik_exe: D:\WILLIAM\ilastik-1.4.1.post1\ilastik.exe
ilastik.project_path: D:\WILLIAM\PHYGEO2026\MORPHO\pore_segmentation\MyProject.ilp
ilastik.n_threads: 4
ilastik.ram_mb: 4096
ilastik.pore_class_index: 0
preprocessing.tile_size: 0
preprocessing.normalization: percentile
preprocessing.norm_percentile_low: 1
preprocessing.norm_percentile_high: 99
postprocessing.threshold_method: otsu
postprocessing.threshold_value: 0.5
postprocessing.morphology_opening_radius: 2
postprocessing.morphology_closing_radius: 2
postprocessing.morphology_fill_holes: True
postprocessing.min_pore_area_px: 50
postprocessing.max_pore_area_px: 0
output.output_dir: results
output.export_formats: ['csv', 'excel']
output.generate_overlay: True
output.generate_report: True
output.overlay_colormap: jet
output.figure_dpi: 150
logging.level: INFO
```

Resumo da Segmentacao

Total de poros detectados : 70

Porosidade total : 29.1548%

Area media dos poros : 4916.0 px

Area mediana dos poros : 386.0 px

Maior poro : 102497 px

Menor poro : 52 px

Metricas PARTISAN - medias:

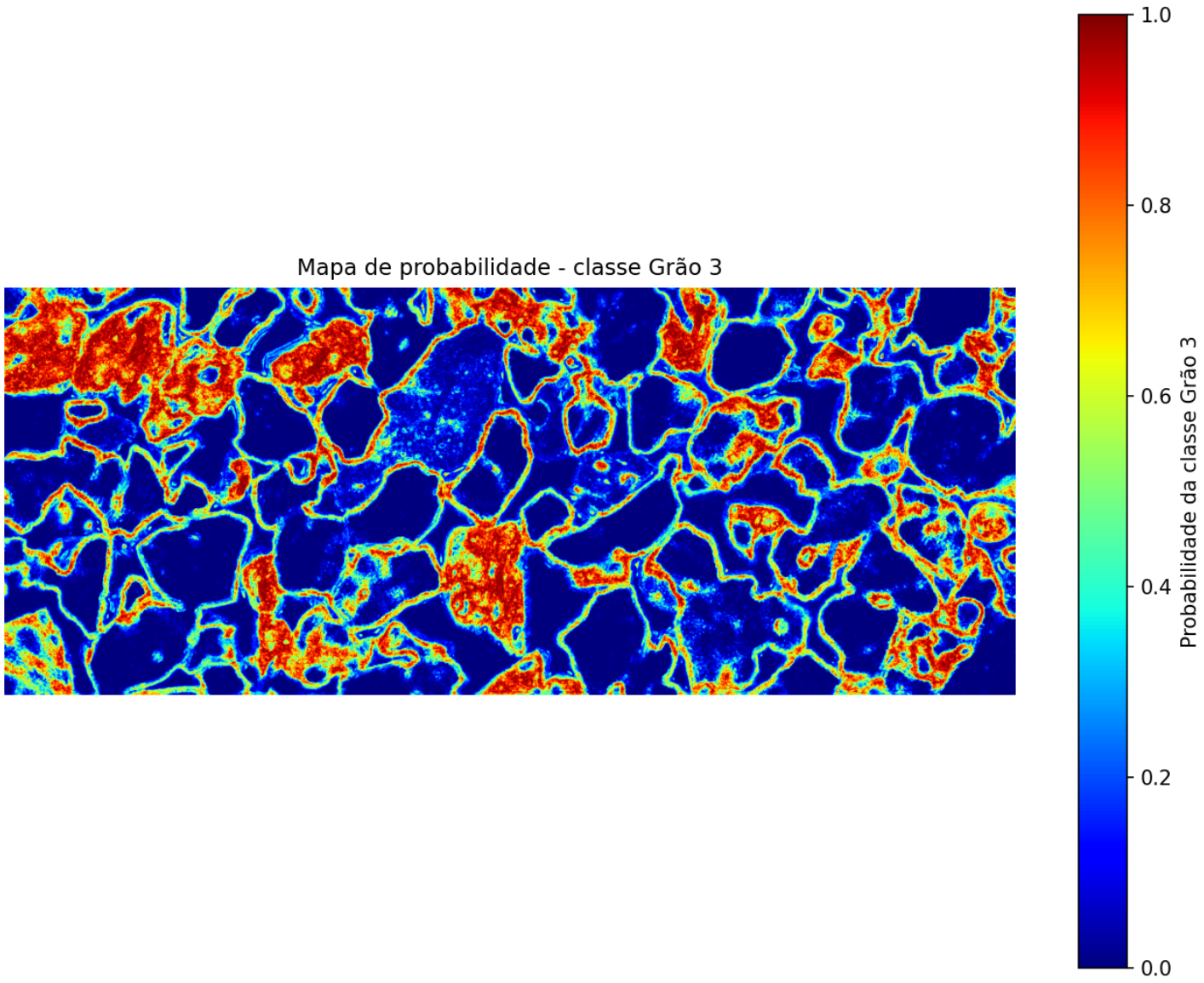
CI_Circ (circularidade) : 0.4647

CI_AR (razao aspecto) : 2.6373

CI_Sol (solidez) : 0.6888

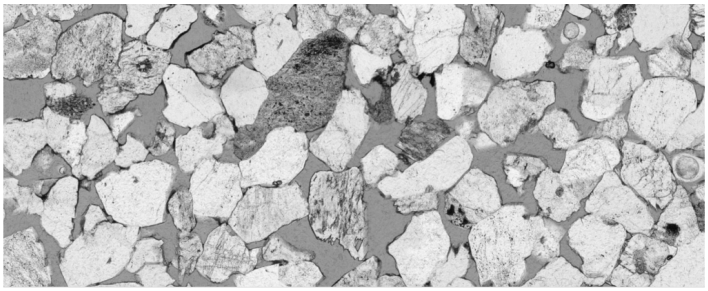
FF (form factor) : 0.4647

Probabilidade

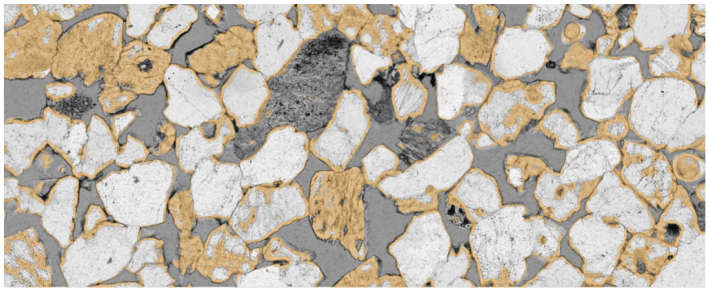


Overlay

Imagem original

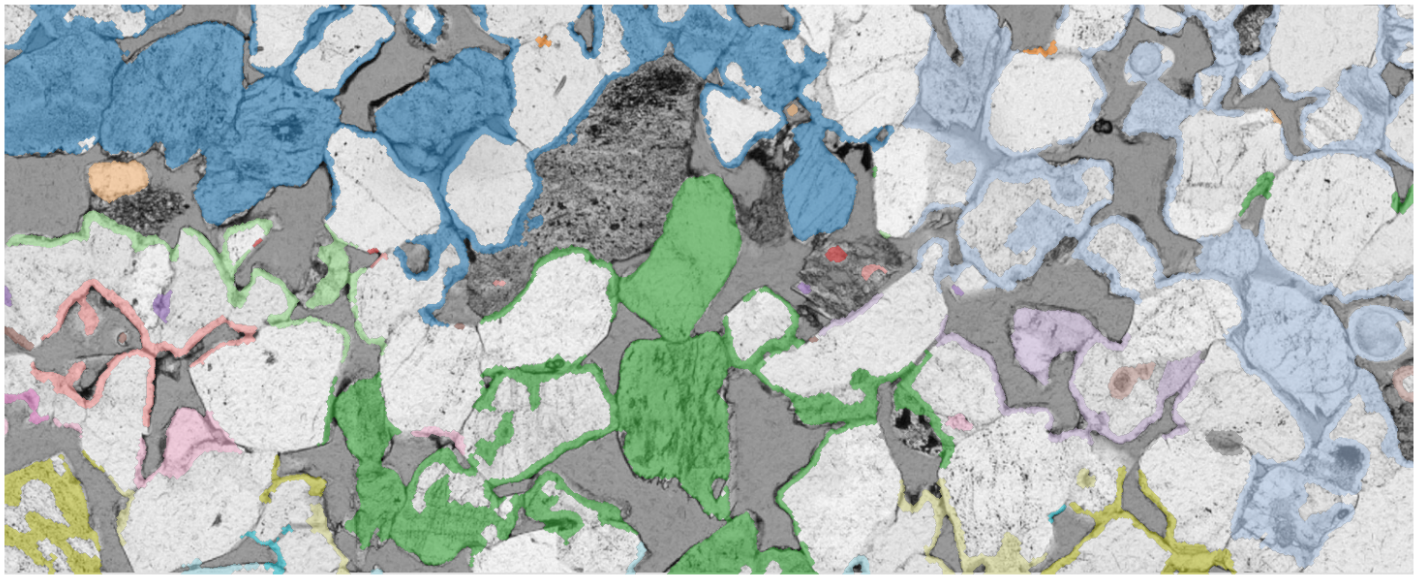


Classe Grão 3 segmentada - fracao: 29.15%

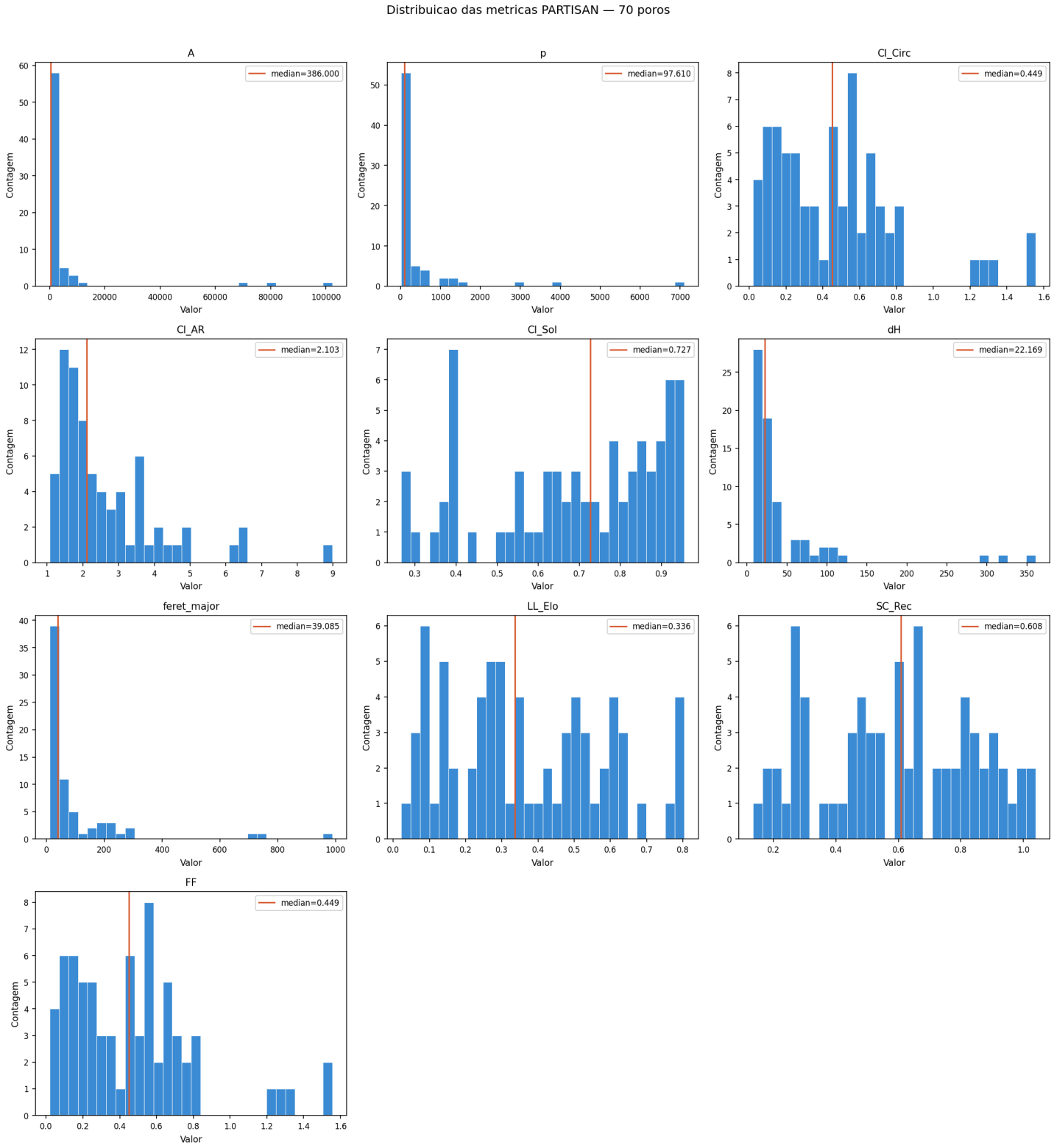


Labels

Poros individuais (N=125)



Histogramas



Estatisticas Descritivas - Metricas PARTISAN

Metrica	Count	Mean	Std	Median	Min	Max
p	70	422.9946	1032.8889	97.6102	22.7279	7098.7201
A	70	4876.0571	17234.8130	386.0000	43.0000	102392.0000
w	70	47.7817	83.2427	17.6924	4.0000	458.9483
b	70	95.0649	165.8358	37.8174	7.7782	984.0000
c	70	136.5532	207.9552	69.6463	23.2455	1134.3262
a	70	48.4143	58.8684	32.0000	8.0000	389.0000
m	70	22.5773	36.4747	9.6442	3.3684	231.0836
Lb	70	93.4000	166.6470	34.5000	10.0000	985.0000
Wb	70	58.0571	95.7433	22.0000	4.0000	518.0000
dH	70	43.4662	66.1942	22.1691	7.3993	361.0672
pHull	70	206.3842	305.5752	85.3614	22.7279	1721.1799
Ahull	70	12861.0000	46954.2468	485.0000	45.0000	265449.0000
ece	70	262.8397	443.7411	101.6109	28.9459	2394.5600
L_maj	70	100.7394	185.2153	39.9728	7.8324	1123.6458
L_min	70	42.4917	69.5595	16.3699	4.5478	399.8617
d_BC	70	98.6971	170.4745	38.0846	11.0000	988.9659
feret_major	70	99.3456	170.1977	39.0845	12.0000	989.6711
feret_minor	70	47.7817	83.2427	17.6924	4.0000	458.9483
w_f	70	95.0649	165.8358	37.8174	7.7782	984.0000
DL_Circ	70	1.9409	1.1196	1.4918	0.8014	7.1447
DL_Rec	70	1.0666	0.3667	1.0127	0.5165	3.1795
DL_Com	70	0.5928	0.2486	0.6083	0.1351	1.0392
DL_Elo	70	3.1218	1.9499	2.5179	0.8098	11.4094
CI_Circ	70	0.4647	0.3424	0.4495	0.0196	1.5569
CI_AR	70	2.6373	1.4971	2.1027	1.0743	8.9851
CI_Con	70	0.9478	0.2739	0.9309	0.2752	1.9952
CI_Sol	70	0.6888	0.2110	0.7268	0.2665	0.9556
LL_Circ	70	0.6346	0.2508	0.6704	0.1400	1.2478
LL_AR	70	0.6342	0.2164	0.6637	0.1964	0.9776
LL_Elo	70	0.3658	0.2164	0.3363	0.0224	0.8036
LL_Con	70	0.8321	0.2025	0.8997	0.2425	1.3876
LL_Sol	70	0.6888	0.2110	0.7268	0.2665	0.9556
LI_AR	70	0.4748	0.1991	0.4756	0.1113	0.9308
LI_Con	70	0.8321	0.2025	0.8997	0.2425	1.3876
LI_Sol	70	0.6888	0.2110	0.7268	0.2665	0.9556
SC_Circ	70	0.3469	0.1890	0.3123	0.0530	0.7691
SC_Rec	70	0.5928	0.2486	0.6083	0.1351	1.0392
SC_ARF	70	0.5265	0.2162	0.5000	0.1426	1.0000
SC_AR	70	2.3186	1.1902	2.0000	1.0000	7.0103
FF	70	0.4647	0.3424	0.4495	0.0196	1.5569
Reg	70	0.1549	0.1773	0.0871	0.0006	0.8487